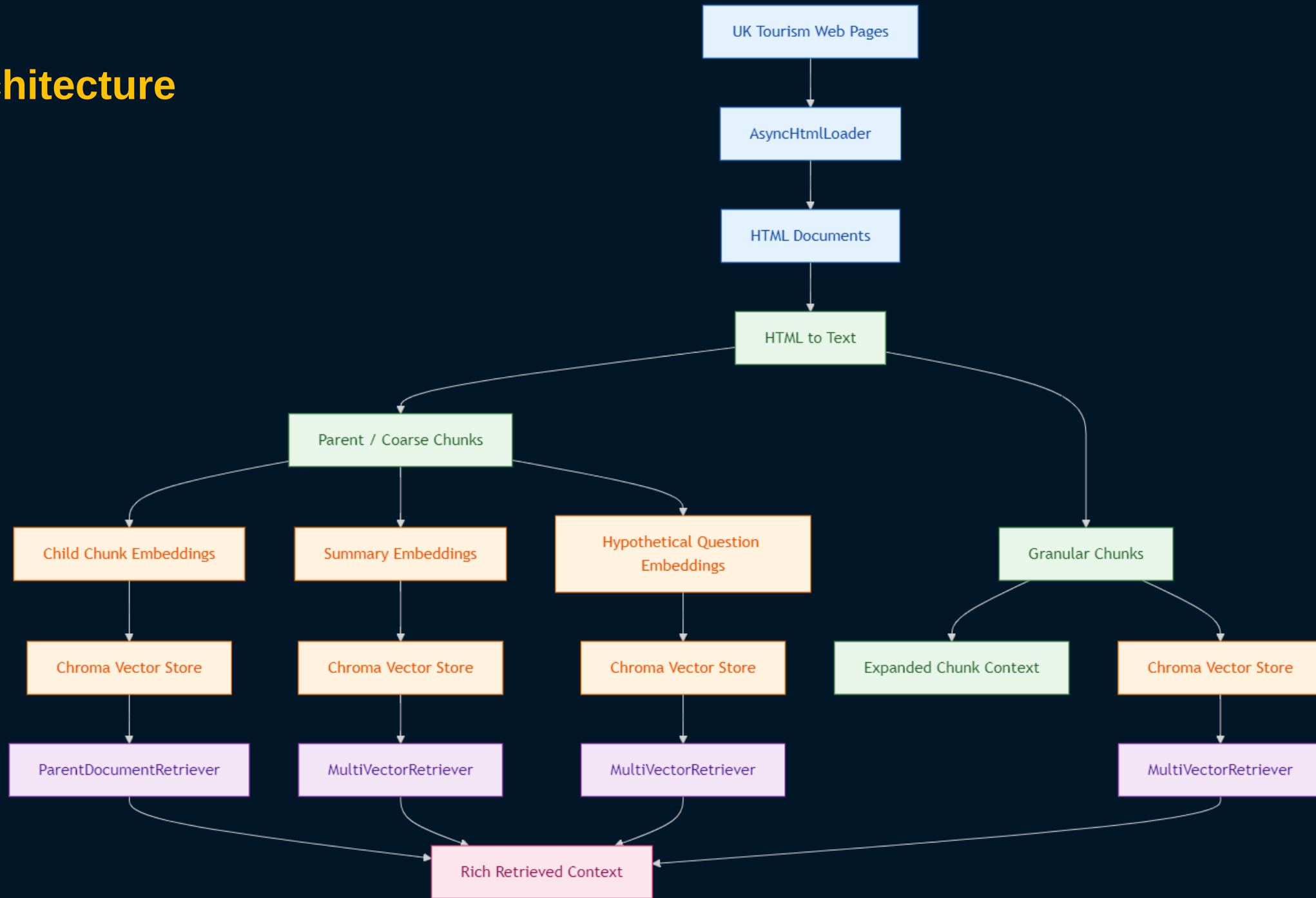


Agentic AI Engineering

Advanced Multi-Vector RAG Embedding Strategies with ChromaDB

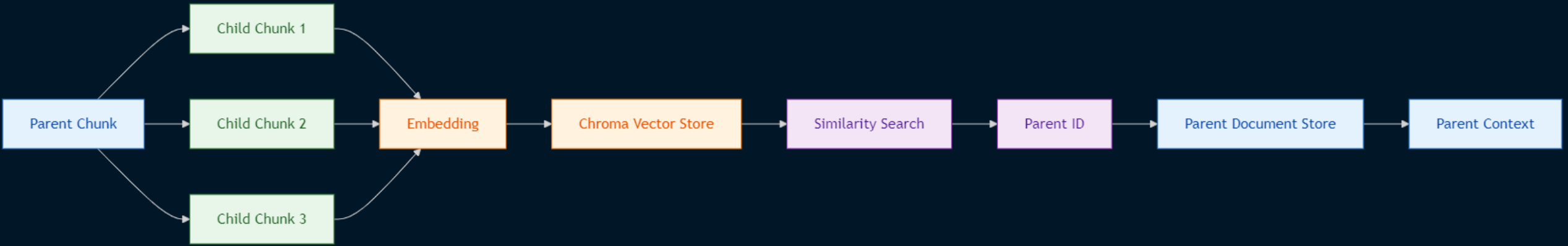
Max Montaseri

Architecture



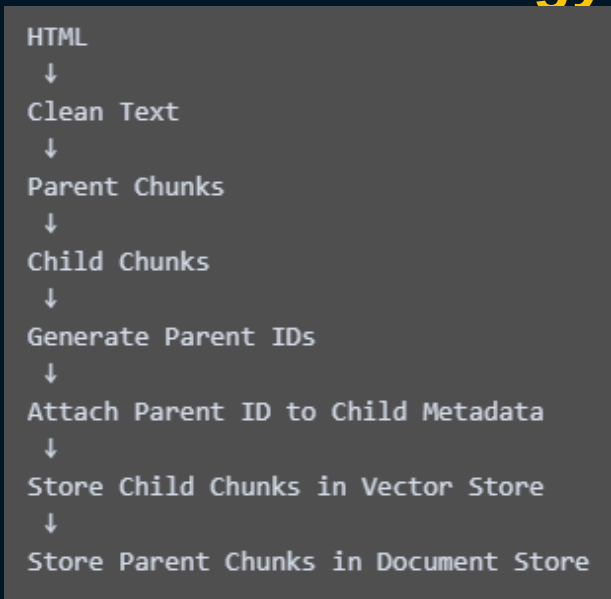
ParentDocumentRetriever Strategy

Data architecture

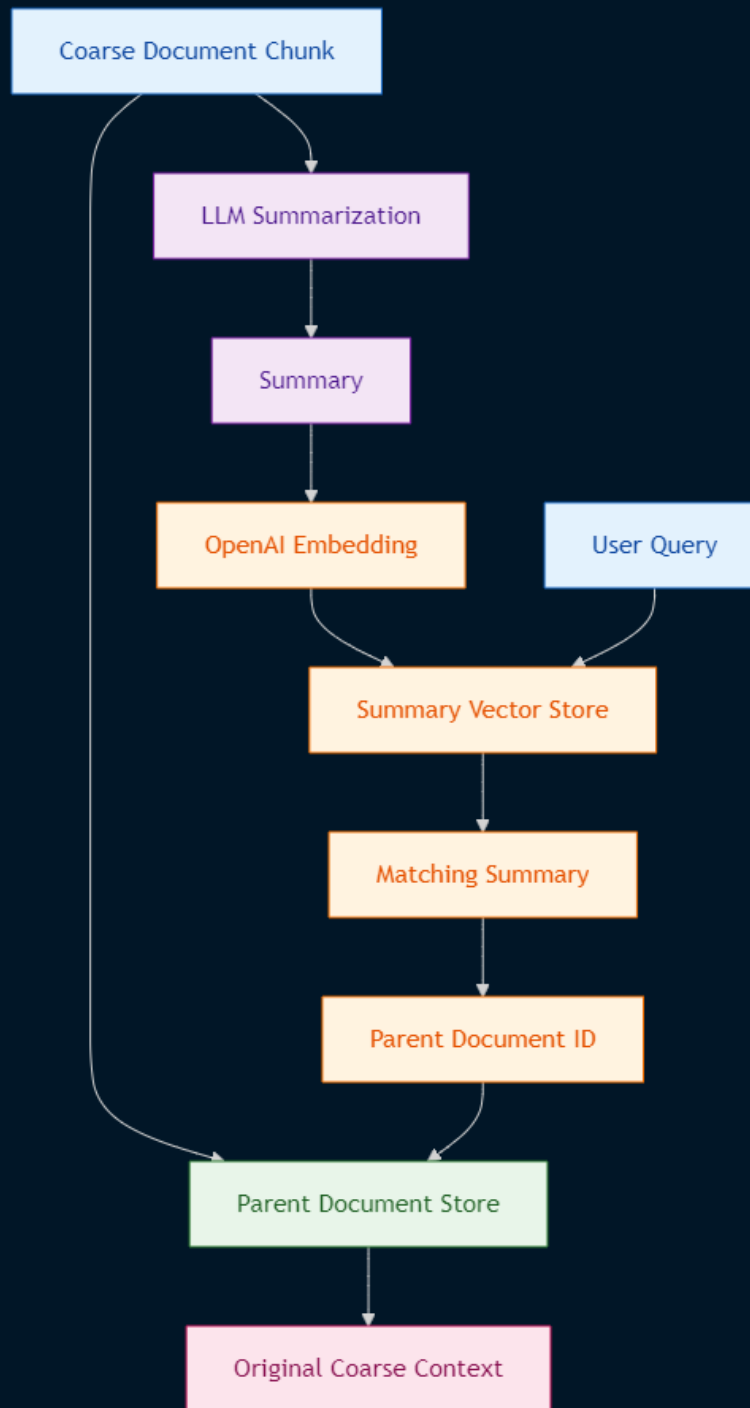


MultiVectorRetriever with Child Chunks Strategy

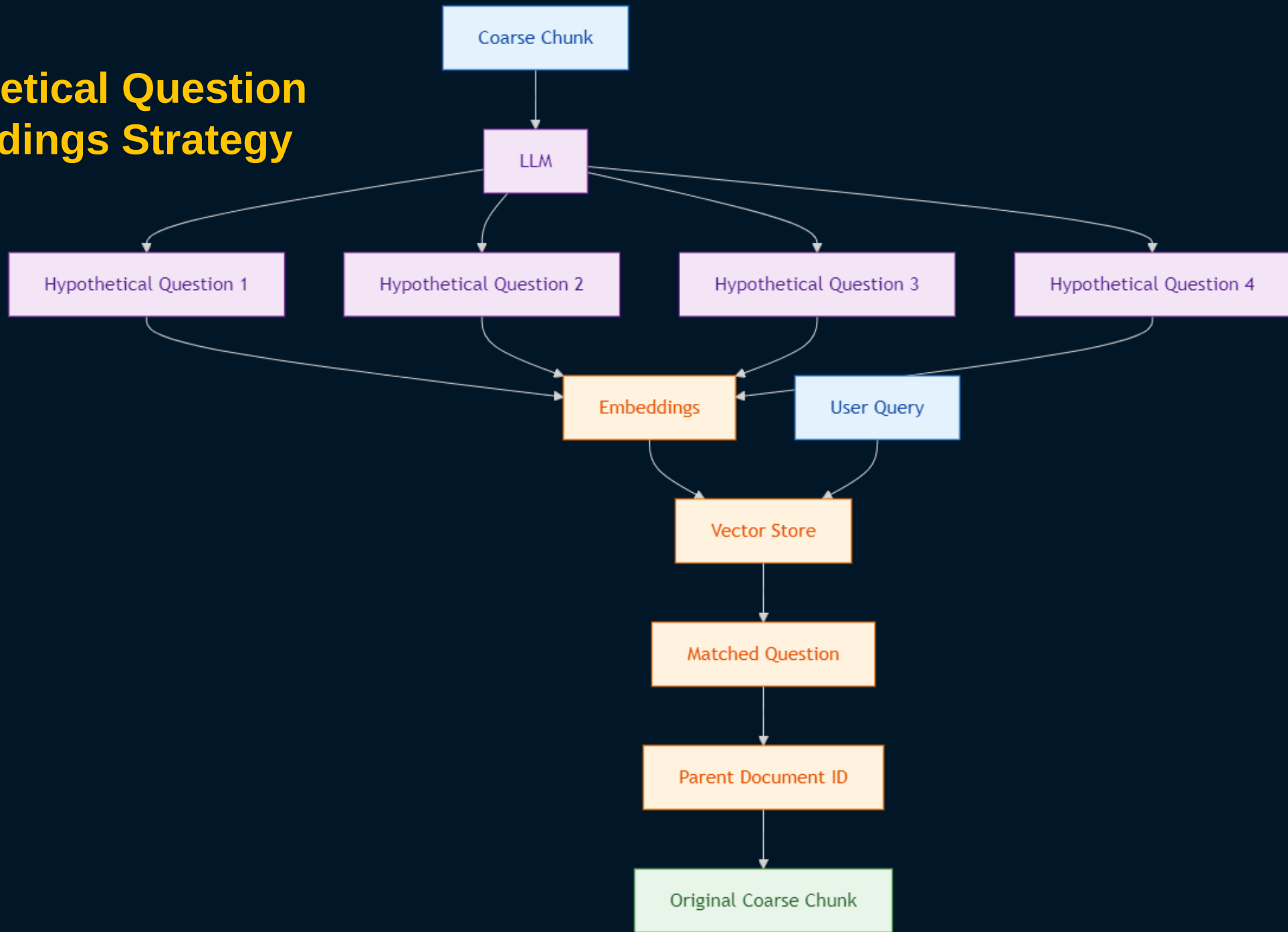
Ingestion flow



Summary Embeddings Strategy



Hypothetical Question Embeddings Strategy



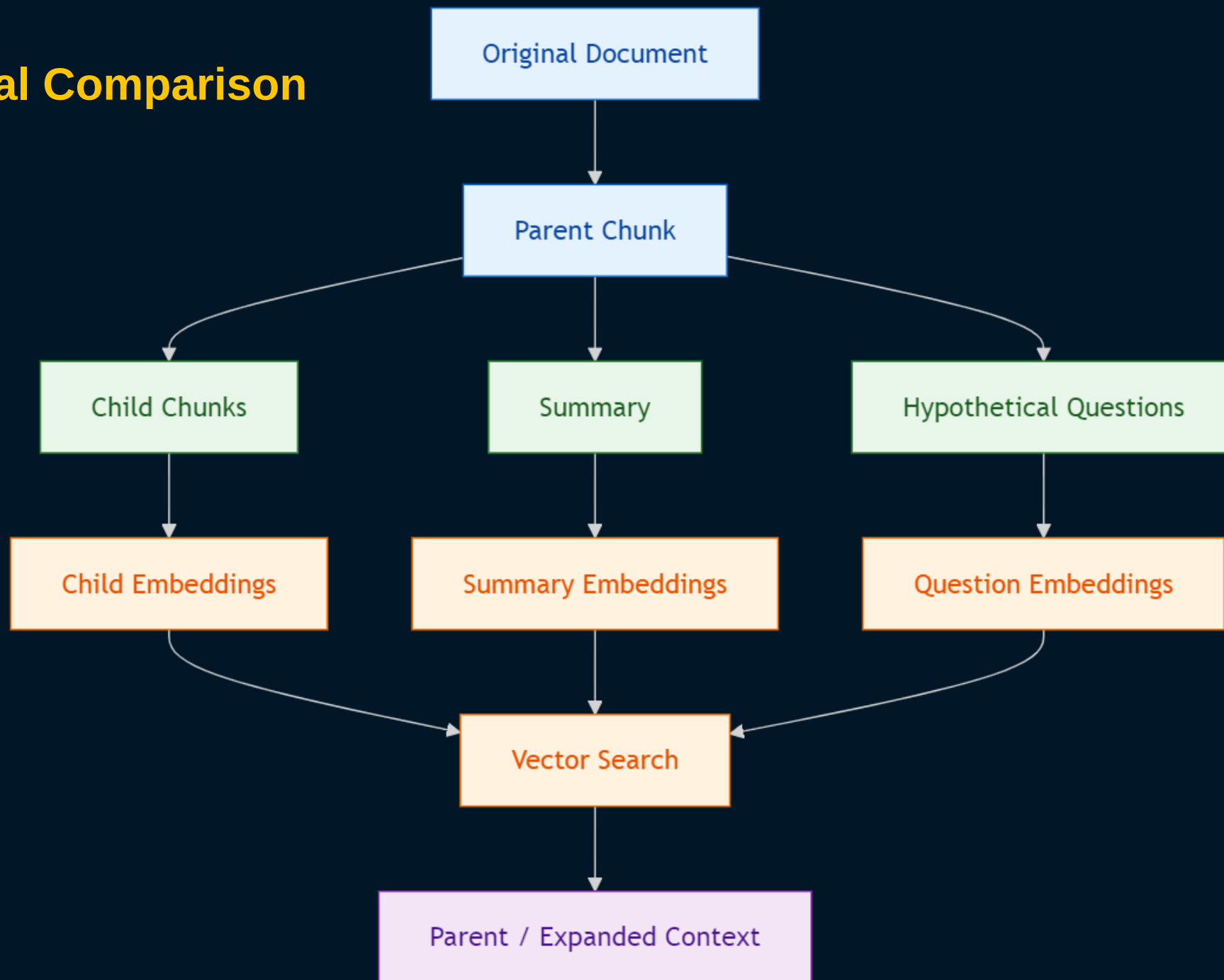
Granular Chunk Expansion Strategy



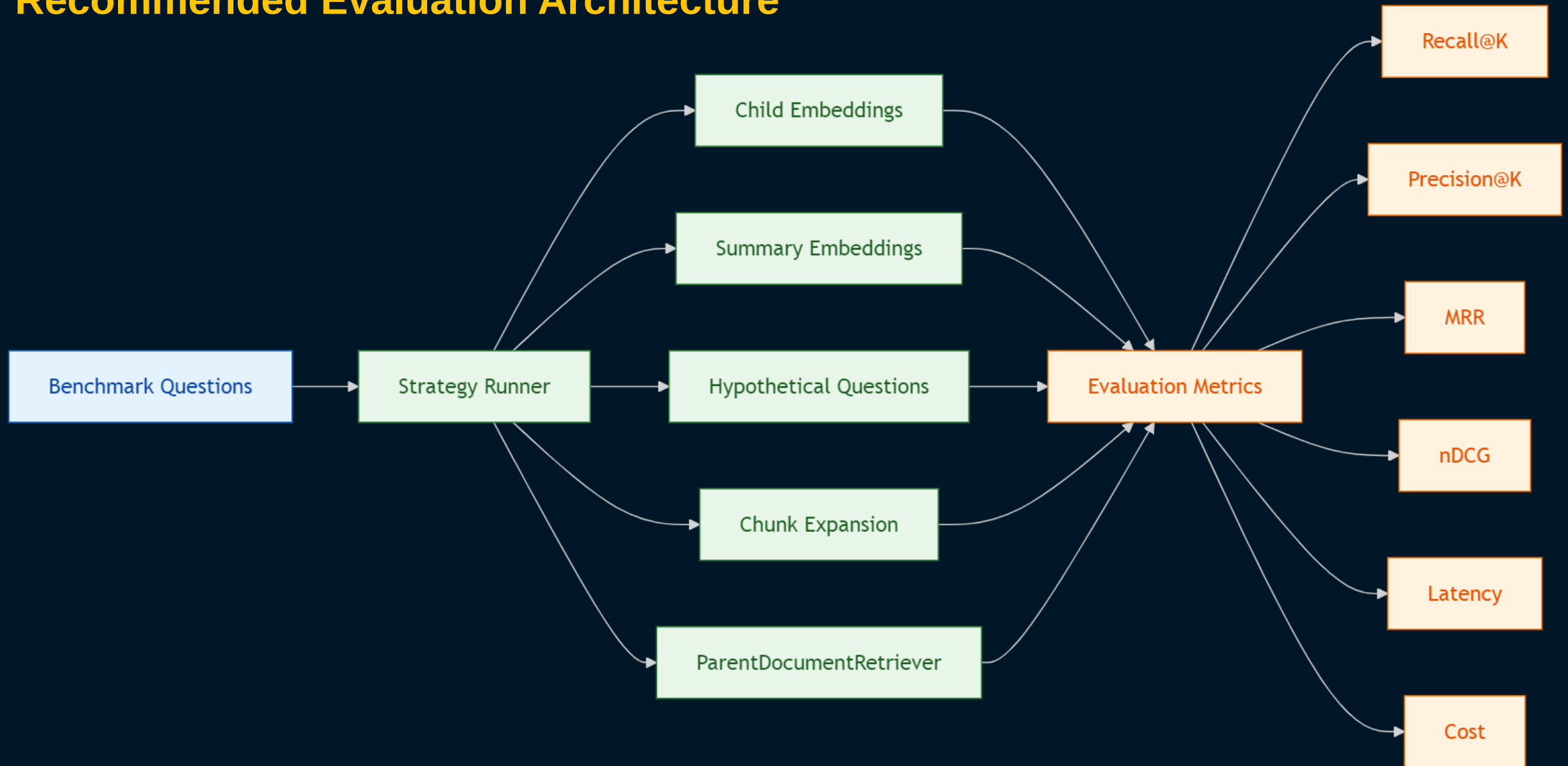
Strategy Comparison

Strategy	Retrieval Representation	Returned Context	Main Strength
ParentDocumentRetriever	Child chunks	Parent chunk	Precise retrieval + broader context
MultiVector + child	Child chunks	Parent chunk	Flexible multi-vector architecture
Summary embeddings	LLM summaries	Parent chunk	Focused semantic representation
Hypothetical questions	Generated questions	Parent chunk	Better query-to-document alignment
Chunk expansion	Current granular chunk	Previous + current + next	Local contextual enrichment

Conceptual Comparison



Recommended Evaluation Architecture



Enterprise RAG Architecture

